

Pre-A Q3LOCK Fixed-Volume Ritz Third Order and Compact-Cylinder Boundary Checkpoint

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This R-167-only gate-level checkpoint synthesizes EXP-000825 and records reusable result R-167 v2.5 without changing a claim card or tier. The result is additive, T0, and `claim_bearing: false`. Its stable result identifier is

[PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#).

R-167 v2.5 preserves every v2.4 and earlier theorem, boundary, negative result, and issued source/PDF pair. It closes exactly one further scoped child. At every fixed finite spatial volume Λ and fixed finite parity-preserving onsite Ritz cutoff M , two sequential block rotations give the complete third-order low-block coefficient

$$\Theta^{(3)} = T^* RCRT - \frac{1}{2}\{A, T^* R^2 T\}. \quad (1.1)$$

The ordered edge-triple expansion cancels a retained disconnected spectator exactly. Every nonzero union is connected, has at most four vertices and diameter at most three, and satisfies a conservative finite-volume/Ritz QPS bound. This is an actual fixed- Λ , fixed- M coefficient theorem, not a spatial-volume-uniform or cutoff-uniform statement.

Separately, for nonzero coupling c , the canonical one-site compact-cylinder split bond subflow fails point-norm C_0 on every nonzero compact cylinder. The unitized compact algebra does not contain the cylinder $K \otimes I$. The compact ideal multiplier algebra contains it but its canonical split-bond action has a norm jump at every nonzero time. This is a scoped carrier/action no-go, not a theorem that a common alpha cannot exist.

The contemporaneous EXP-000820 and EXP-000821 records repair authority linkage only. They add no theorem or reusable-result increment. The records EXP-000822, EXP-000823, and EXP-000824 are T0 strategy and origin-hypothesis records only; they prove no common dynamics, emergent time, propagation geometry, gravity, or physical spacetime. The theorem-bearing exploration for this checkpoint is only EXP-000825.

R-168 v1.3 remains historical and is not reissued. No R-168 verification count, script hash, theorem, or lifecycle metadata is part of this document.

2 Closed child, negative authority, and surviving parents

The one child closed by EXP-000825 is exactly

PA-CP1-ST8-Q3LOCK-FIXED-FINITE-VOLUME-AND-RITZ-COMPLETE-THIRD-ORDER-LINKED-RANK-TWO-LOW-BLOCK-COEFFICIENT.

Its scope is every separately fixed finite Λ and fixed finite parity-preserving onsite Ritz cutoff M . The one new negative authority is exactly

NG-2026-08-12-PRE-A-ST8-Q3LOCK-CANONICAL-ONE-SITE-COMPACT-CYLINDER-BOND-SUBFLOW-POINT-NORM-CO.

The following load-bearing parents and successors remain **OPEN**:

1. PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA;
2. PA-CP1-ST8-Q3LOCK-DLR-TO-COMMON-ALPHA-KMS-IDENTIFICATION;
3. PA-CP1-ST8-Q3LOCK-CONNECTED-RANK-TWO-OSCILLATOR-ELIMINATION-QPS-NORM-AND-CUTOFF-COMPATIBILITY;
4. PA-CP1-ST8-Q3LOCK-INFINITE-DIMENSIONAL-RANK-TWO-BAND-BLOCK-DIAGONALIZATION-AND-TWO-PHASE-QPS;
5. PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY;
6. PA-CP1-ST8-Q3LOCK-A-TO-ZERO-ENLARGED-COUNTERTERM-REGULAR-CONTINUUM;
7. PA-CP1-ST8-Q3LOCK-PHYSICAL-EMPTY-SPACE-REFERENCE;
8. PA-PRE-A-CO-N1-N5-VALIDATION;
9. PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE.

Neither the positive coefficient theorem nor the negative carrier boundary closes any parent in this list.

3 Setting and two sequential rotations

Fix one finite spatial volume Λ and one finite parity-preserving onsite Ritz cutoff M . Every block below is therefore a bounded finite matrix. Relative to a rank-two low projection P and $Q = 1 - P$, write

$$H(\lambda) = K + \lambda(V_d + V_{od}), \quad V_d = A \oplus C, \quad T = QVP. \quad (3.1)$$

Normalize the low energy so that $PKP = 0$, assume $QKQ \geq \Gamma Q$, and put

$$R = (QKQ)^{-1}, \quad D = RT, \quad G = D - D^*. \quad (3.2)$$

The inverse is extended by zero on P . The retained parity hypotheses say that K , P , and V commute with global parity. Consequently the objects in both rotations preserve the declared parity sectors. Direct block multiplication gives

$$[G, K] = -V_{od}. \quad (3.3)$$

For the first rotation,

$$H_1(\lambda) = e^{\lambda G} H(\lambda) e^{-\lambda G}, \quad (3.4)$$

collecting powers through order three gives

$$H_1(\lambda) = K + \lambda V_d + \lambda^2 \left([G, V_d] + \frac{1}{2} [G, V_{od}] \right) + \lambda^3 \left(\frac{1}{2} [G, [G, V_d]] + \frac{1}{3} [G, [G, V_{od}]] \right) + O(\lambda^4). \quad (3.5)$$

The remaining second-order off-diagonal block is removed by defining

$$Z = R(DA - CD), \quad G_2 = Z - Z^*. \quad (3.6)$$

The QP block of $[G, V_d]$ is $DA - CD$, and another direct block multiplication gives

$$[G_2, K] = -[G, V_d]. \quad (3.7)$$

Apply the second rotation

$$H_2(\lambda) = e^{\lambda^2 G_2} H_1(\lambda) e^{-\lambda^2 G_2}. \quad (3.8)$$

Equation (3.7) cancels the remaining order-two off-diagonal block. At order three, $[G_2, V_d]$ is off diagonal. Also G and V_{od} are off diagonal, so $[G, V_{od}]$ is block diagonal and $[G, [G, V_{od}]]$ is off diagonal. Projection therefore leaves the complete low block

$$\Theta^{(3)} = \frac{1}{2} P[G, [G, V_d]] P. \quad (3.9)$$

Using $D = RT$, exact multiplication yields

$$\boxed{\Theta^{(3)} = T^* R C R T - \frac{1}{2} \{A, T^* R^2 T\}.} \quad (3.10)$$

For machine-readable reproduction, the same formula is denoted $\text{Theta} = T^* R C R T - (1/2) \{A, T^* R^2 T\}$. Equation (3.10) is the complete coefficient for these two specified sequential rotations. It is not a selected diagram, a scalar-denominator expansion, a fourth-order statement, or an all-order convergence theorem.

4 Linked edge triples and conservative QPS control

Resolve the local blocks as $T = \sum_e T_e$, $A = \sum_f A_f$, and $C = \sum_f C_f$. The ordered (e, f, g) term is

$$\Theta_{efg}^{(3)} = T_e^* R C_f R T_g - \frac{1}{2} \left(A_f T_e^* R^2 T_g + T_e^* R^2 T_g A_f \right). \quad (4.1)$$

If the retained edge f is disconnected from $e \cup g$, its high-block operator C_f is the same retained spectator as A_f on the exact high-support sector created by T_g . It commutes with R , T_e , and T_g , so

$$T_e^* R C_f R T_g = A_f T_e^* R^2 T_g = T_e^* R^2 T_g A_f. \quad (4.2)$$

Substitution into (4.1) gives exact cancellation. If another component is disconnected, the onsite high-support sectors make the corresponding $T_e^* \cdots T_g$ block orthogonal. Thus every nonzero ordered triple has connected union. A connected union of three edges has at most four vertices and diameter at most three, including repeated-edge cases.

At this fixed (M, Λ) , define

$$\rho_{M, \Lambda} = \sup_f \|R^{1/2} C_f R^{1/2}\|, \quad a_{M, \Lambda} = \sup_f \|A_f\|. \quad (4.3)$$

If $\|T_e\| \leq \epsilon$ and $\|R\| \leq 1/\Gamma$, then

$$\|\Theta_{efg}^{(3)}\| \leq \epsilon^2 \left(\frac{\rho_{M,\Lambda}}{\Gamma} + \frac{a_{M,\Lambda}}{\Gamma^2} \right). \quad (4.4)$$

For maximum graph degree z , a safe rooted ordered count is

$$N_{3,\text{root}} \leq 12z(2z-1)^2. \quad (4.5)$$

The factor is deliberately conservative: an oriented spanning walk gives at most $z(2z-1)^2$ choices, followed by six orderings and two safe root incidences. Since $|X| \leq 4$ and $\text{diam}(X) \leq 3$, the QPS interaction norm obeys

$$\boxed{\|\Theta^{(3)}\|_a \leq 48z(2z-1)^2 e^{3a} \epsilon^2 \left(\frac{\rho_{M,\Lambda}}{\Gamma} + \frac{a_{M,\Lambda}}{\Gamma^2} \right)}. \quad (4.6)$$

No optimality is asserted. The combinatorial factor is independent of Λ , but $\rho_{M,\Lambda}$ and $a_{M,\Lambda}$ are only fixed-volume/Ritz quantities. There is no Lambda-uniform estimate, no spatial-volume or thermodynamic limit, and no third-order Ritz-tail τ_M estimate.

5 Numerical verification and exact matrix fixtures

For the scalar two-block fixture, take

$$\Gamma = 2, \quad \epsilon = \frac{1}{10}, \quad A = \frac{1}{3}, \quad C = \frac{5}{3}. \quad (5.1)$$

Then $D = 1/20$, $Z = -1/30$, and both (3.9) and (3.10) give

$$\Theta^{(3)} = \frac{1}{300}. \quad (5.2)$$

The pure-offdiagonal limit $A = C = 0$ gives $Z = 0$ and $\Theta^{(3)} = 0$, as required by block grading. Because this scalar check cannot expose an ordering error, the second fixture is noncommutative. Use

$$K_Q = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 1 & 0 \\ 1 & -2 & 2 \\ 0 & 2 & 4 \end{pmatrix}, \quad (5.3)$$

$$T = \frac{1}{10} \begin{pmatrix} 1 & 2 \\ 0 & -1 \\ 3 & 1 \end{pmatrix}, \quad R = K_Q^{-1}. \quad (5.4)$$

Exact rectangular multiplication gives

$$D = \begin{pmatrix} 1/20 & 1/10 \\ 0 & -1/30 \\ 3/50 & 1/50 \end{pmatrix}, \quad Z = \begin{pmatrix} 1/20 & -2/15 \\ -71/900 & -13/225 \\ -7/250 & 13/750 \end{pmatrix}. \quad (5.5)$$

Embedding the blocks into $P \oplus Q$ checks the sign in (3.7), the second-offdiagonal cancellation, and the two projected discarded terms

$$P[G_2, V_d]P = 0, \quad P[G, [G, V_{\text{od}}]]P = 0. \quad (5.6)$$

Both the nested commutator and block formula then give

$$\Theta^{(3)} = \begin{pmatrix} 17/5000 & -313/90000 \\ -313/90000 & 431/22500 \end{pmatrix}. \quad (5.7)$$

This exact nonscalar fixture checks ordering and signs without simultaneous diagonalization. It is still one fixed finite Λ and M .

The retained v2.1 rational fixture has

$$\rho_{M,\Lambda} = \frac{15201}{156250}, \quad \epsilon = \frac{23}{6250}, \quad \Gamma = 100, \quad a_{M,\Lambda} = \frac{96139}{1500000}, \quad z = 6, \quad e^a = 2. \quad (5.8)$$

The per-triple bound (4.4) is

$$\frac{7770533371}{585937500000000000}, \quad (5.9)$$

and the safe QPS value from (4.6) is

$$\frac{2820703613673}{762939453125000}. \quad (5.10)$$

For exact ASCII verification, the scalar fixture token is 1/300 and the safe QPS token is 2820703613673/762939453125000. Their scope includes **no spatial-volume or thermodynamic limit**.

The primary symbolic engine constructs both exact block forms. The non-importing independent engine uses standard-library rational arithmetic and imports neither the primary module nor a primary result file.

6 Canonical compact-cylinder point-norm obstruction

Let

$$\mathcal{H}_x = \mathcal{H}_y = L^2(\mathbb{R}^8), \quad B_\delta = \exp\left(\frac{ic\delta}{\hbar} q_x \cdot q_y\right), \quad \beta_\delta = \text{Ad}(B_\delta), \quad (6.1)$$

where $c \neq 0$ and $\hbar > 0$. For a nonzero compact operator K on $\mathcal{H}_x$, the formal one-site cylinder is $A = K \otimes I$. Direct-integral decomposition over q_y gives, for every $\delta \neq 0$,

$$\begin{aligned} \|\beta_\delta(A) - A\| &= \text{ess sup}_{r \in \mathbb{R}^8} \|M_{c\delta r/\hbar}^* K M_{c\delta r/\hbar} - K\| \\ &= \sup_{s \in \mathbb{R}^8} \|M_s^* K M_s - K\|. \end{aligned} \quad (6.2)$$

The modulations M_s converge weakly to zero as $|s| \rightarrow \infty$. Compactness gives $K M_s \xi \rightarrow 0$ for each fixed vector ξ , and hence

$$\sup_s \|M_s^* K M_s - K\| \geq \|K\|. \quad (6.3)$$

Thus the multiplier action has a nonzero norm jump at every nonzero time. For an exact rank-one fixture, set

$$\psi(q) = 2^{-4} \exp\left[-\frac{1}{2} \sum_{j=1}^8 |q_j|\right], \quad K = P_\psi. \quad (6.4)$$

The vector is normalized and

$$\langle \psi, M_s \psi \rangle = \prod_{j=1}^8 \frac{1}{1 + s_j^2}. \quad (6.5)$$

The exact supremum in (6.2) is one. With $c = \hbar = 1$, $\delta_n = 1/n$, and $r_n = (n/2)e_1$, the modulation is $s = e_1/2$, so

$$|\langle \psi, M_s \psi \rangle| = \frac{4}{5}, \quad \|P_{M_s \psi} - P_\psi\| = \frac{3}{5}, \quad \|P_{M_s \psi} - P_\psi\|^2 = \frac{9}{25}. \quad (6.6)$$

The unitized compacts on $\mathcal{H}_x \otimes \mathcal{H}_y$ do not contain $K \otimes I$: its infinite $\mathcal{H}_y$ multiplicity prevents compactness modulo a scalar. The multiplier of the nonunital compact ideal is

$$M(\mathcal{K}(\mathcal{H}_x \otimes \mathcal{H}_y)) = B(\mathcal{H}_x \otimes \mathcal{H}_y). \quad (6.7)$$

It contains the cylinder, but (6.2)–(6.3) show that its canonical split-bond action is not point-norm C_0 . This compact ideal multiplier boundary is different from the older raw momentum Weyl/basic-resolvent obstruction. The older route tested a raw momentum Weyl or one basic bounded momentum resolvent of the unbounded p_x ; the new theorem tests every nonzero one-site compact cylinder and the unitized-compact versus multiplier dichotomy.

The conclusion is restricted to this canonical compact-cylinder split bond carrier/action pair. It does not reject an unsplit dynamics, another local-strict carrier, state-weighted convergence, orbit-smearing, or a common alpha obtained by another route.

7 Devil's-advocate review

1. **DISMISSED – a third-order BCH term was omitted:** equation (3.5) contains both nested order-three terms. The V_{od} term and the second-rotation $[G_2, V_d]$ term are off diagonal after the relevant projection, leaving exactly (3.9).
2. **DISMISSED – the second generator has the wrong sign:** the QP block of $[G, V_d]$ is $DA - CD$; multiplication by R and direct block calculation give (3.7).
3. **DISMISSED – the anticommutator has a missing factor:** the outer factor $1/2$ in (3.9) produces one T^*RCRT term and precisely $-(1/2)\{A, T^*R^2T\}$.
4. **DISMISSED – the scalar fixture can detect ordering:** it cannot. The independent noncommutative rectangular fixture (5.3)–(5.7) checks the sign, discarded terms, and ordering explicitly.
5. **DISMISSED for this coefficient – a retained disconnected spectator survives:** equation (4.2) cancels it exactly in this local unperturbed-resolvent expansion. The v2.4 negative used a different raw global scalar Feshbach denominator and remains valid in its own scope.
6. **UPHELD as an overclaim – the QPS constant is sharp:** the factor $12z(2z-1)^2$ is a deliberate overcount and $|X| \leq 4$ is paid without optimization.
7. **UPHELD as an overclaim – finite matrices give a uniform third-order theorem:** $\rho_{M,\Lambda}$ and $a_{M,\Lambda}$ may grow with either cutoff. There is no Lambda-uniform, cutoff-uniform, unbounded, or thermodynamic theorem here.
8. **UPHELD as an overclaim – finite matrix exponentials close all orders:** the result is one complete coefficient through order three. It gives no fourth coefficient, linked recursion, convergent BCH or Lie–Schwinger series, all-order interaction, or phase transfer.

9. **UPHELD as an overclaim – the compact-cylinder fixture proves common-alpha nonexistence:** it rejects one canonical split-bond carrier/action pair only. Other carriers and unsplit constructions remain open.
10. **DISMISSED – 4/5 was converted using vector distance:** equation (6.6) uses $\|P_u - P_v\|^2 = 1 - |\langle u, v \rangle|^2$, yielding 9/25 and projection distance 3/5.
11. **DISMISSED – the new no-go duplicates the old resolvent fixture:** the older fixture tested one bounded function of momentum. This one tests every nonzero compact cylinder and the compact-ideal multiplier boundary.
12. **VALID with mitigation – passing verification closes physical parents:** the engines bind only the formulas, authorities, lifecycle, and scope declared here. Their passing totals change no claim tier and close no open parent.

Independent adversarial review is invited especially on the two rotation signs, the coefficient 1/2, both projected discarded terms, exact spectator cancellation, support counting, the conservative QPS payment, noncommutative rational multiplication, the direct-integral essential supremum, compactness argument, projection-distance identity, and every scope firewall.

8 Reproduction and exact verification pins

The R-167 v2.5 primary, non-importing independent, and integrated scripts are

1. [codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split.py](#);
2. [codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_independent.py](#);
3. [codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_verify.py](#).

From the repository root, run

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$r167 = (
    'codes\foundations\pre_a_cp1_st8_q3lock_' +
    'local_measured_renyi_semiclassical_doublet_route_split'
)
& $py -X utf8 ($r167 + '.py')
& $py -X utf8 ($r167 + '_independent.py')
& $py -X utf8 ($r167 + '_verify.py')
```

The expected final totals are primary PASS 384/384, non-importing independent PASS 218/218, and integrated PASS 369/369. The exact raw verification-script SHA-256 pins are

```
R-167 primary:
9d667e615879985fa32dfb9540109769cd830e53aebb8c226f8b3a1008dea94c
R-167 independent:
f3e8bc777a2756f67168b3afea72f8a2365a7efc7f1a78e8b6ac3e5add3a7b71
R-167 integrated:
6a3ccc658817fdf715509e80acf8dd993691019d7db455942811cbe63389ae5a
```

The independent pin is the raw script SHA-256, not the canonical stdout payload hash. The primary and independent engines recompute the third-order and compact-cylinder fixtures from labelled inputs. The integrated engine binds those computations to EXP-000825, R-167 v2.5, the exact gate and negative authorities, the EXP-000820/EXP-000821 linkage-only boundary, the EXP-000822–EXP-000824 T0-only boundary, generated surfaces, stored runs, and checkpoint lifecycle. A missing or failed check invalidates the total.

This document is R-167-only. R-168 v1.3 remains historical and is not reissued; no R-168 count or hash appears here. No per-lemma or intermediate PDF was issued. One R-167 v2.5 gate-level synthesis source/PDF pair may be issued only after the primary, non-importing independent, integrated, formal-authority, generated-surface, source-form, freshness, dual-extraction, strict-release, and visual-review checks pass.

9 Exact no-overclaim and dependency boundary

The common-alpha parent is **OPEN**. The compact-cylinder result rejects one canonical split-bond carrier/action pair; it does not prove common-alpha nonexistence. There is no actual all-shape or all-exhaustion Q3 Cauchy estimate, no identified local generator, and no KMS quotient.

The connected oscillator-elimination parent is **OPEN**. The complete third-order coefficient at fixed finite Λ and M gives no spatial-volume or thermodynamic limit, no Λ -uniform constant, no third-order τ_M cutoff estimate, no cutoff-uniform or unbounded theorem, no fourth-order coefficient, no all-order linked recursion, and no convergent BCH or Lie–Schwinger series.

The broader rank-two and broken-sector spectral parents are **OPEN**. No boundary/source-uniform quasi-local elimination, exact two-phase oscillator QPS, phasewise intertwiner, temporal-mass estimate, oscillator-lattice GNS coercivity, broken-sector GNS gap, or physical mass gap follows.

The Round-1 and cross-candidate parents are **OPEN**. EXP-000820 and EXP-000821 only repair authority linkage; EXP-000822–EXP-000824 only register T0 route hypotheses; EXP-000825 changes neither the evidence-role freeze nor the prospective holdout state.

Nothing here proves regulator removal, continuum control, same-Hamiltonian physical-empty comparison, a below-empty sign, physical vacuum selection, emergent time, a universal propagation cone, gravity, C6 emergence, CP1, physical Sector A, or Pre-A closure. The registered A5–SECTOR–A–SYNTHESIS remains conditional at its declared scope, and the C6–SPACETIME–SIGNATURE claim card and tier are unchanged.

10 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:	R-167 v2.5.
Theorem exploration:	EXP-000825.
Authority corrections:	EXP-000820 and EXP-000821; linkage only, no theorem or reusable-result increment.
T0 strategy records:	EXP-000822, EXP-000823, and EXP-000824; no theorem, physical claim, or reusable-result increment.
Precise statement:	At each fixed finite spatial volume Λ and fixed finite parity-preserving onsite Ritz cutoff M , two sequential rotations have complete third-order low block $\Theta = T^* \text{RCRT} - (1/2)\{A, T^* R^2 T\}$; disconnected retained spectators cancel in every ordered edge triple, every nonzero union is connected with support at most four and diameter at most three, and the conservative fixed-volume/Ritz QPS bound follows. For nonzero coupling, the canonical one-site compact-cylinder split bond action fails point-norm C_0 on every nonzero compact cylinder at the compact ideal

multiplier boundary.

Claim context: C6-SPACETIME-SIGNATURE; claim_bearing false.

Closed child: Exactly the fixed-finite-volume-and-Ritz third-order gate in Section 2.

Negative result: Exactly the canonical one-site compact-cylinder split-bond point-norm-C0 no-go in Section 2.

Scope: T0 additive analytic/exact result at separately fixed finite Λ and M , plus one canonical compact-cylinder carrier/action no-go.

Dependencies: R-167 v2.4; EXP-000818; rank-two low/high block and parity setup; positive fixed finite Ritz gap; exact onsite high-support sectors; canonical split bond unitary; compactness of K .

Evidence grade: ANALYTIC + EXACT + primary + non-importing independent + integrated; R-167-only checkpoint; R-168 v1.3 historical and not reissued.

Reproduction command: Run the three commands in Section 8 from repository root, primary then independent then integrated.

Expected output: PASS 384/384; PASS 218/218; PASS 369/369.

Primary raw SHA-256: 9d667e615879985fa32dfb9540109769cd830e53aebb8c226f8b3a1008dea94c.

Independent raw SHA-256: f3e8bc777a2756f67168b3afea72f8a2365a7efc7f1a78e8b6ac3e5add3a7b71.

Integrated raw SHA-256: 6a3ccc658817fdf715509e80acf8dd993691019d7db455942811cbe63389ae5a.

Falsification gate: Any rotation sign/factor, projected-term, exact matrix, spectator cancellation, support/count/QPS, direct-integral, compactness, projection-distance, assertion-count, hash, provenance, or scope mismatch.

Tier before / after: C6 T1 / C6 T1; no claim or tier change.

No-overclaim statement: No spatial-volume or thermodynamic limit, Λ -uniform or cutoff-uniform third order, unbounded or all-order oscillator elimination, QPS phase transfer, common-alpha nonexistence, actual Q3 common alpha, generator/KMS quotient, oscillator GNS gap, Round-1, regulator removal, continuum, physical-empty sign, C6, CP1, physical Sector A, or Pre-A closure.

Next required action: Prove actual bidirectional full-directed-set Q3 point-norm Cauchy control on a viable common carrier; independently obtain volume/cutoff control beyond the fixed third-order coefficient and construct a convergent connected all-order interaction before any phase-wise oscillator/GNS transfer.

PDF policy: No per-lemma or intermediate PDF. One R-167-only v2.5 gate-level pair is issued after all proof, authority, extraction, release, and visual gates.

R-168 firewall: R-168 v1.3 remains historical and is not reissued; no R-168 verification count or hash is part of this document.